

Group No. 11



HieraSync AI

Smart Academic Workflow &
Event Management System

One role-aware workspace for tasks, approvals, events,
notifications, analytics and AI assistance.

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At a Glance & Presentation Roadmap

PROJECT AT A GLANCE

17+ — REST API modules (FastAPI)

13+ — Interactive Web Pages (React 19)

10 — Role-Based User Access Levels

2-Stage — HOD → Principal Approval Engine

Zero-Cost — Heuristic AI Risk & Report Engine

PRESENTATION SEQUENCED ROADMAP

01. Title & Metadata

02. Introduction & Platform Scope

03. Problem Statement &
Objectives

04. Literature Survey &
Comparative Analysis

05. System Design & Architecture

06. Technology Stack &
Justifications

07. Developed Modules & Code
Audit

08. Advantages, ROI &
Applications

09. References & Bibliography

10. Conclusion & Q&A Appendix

Introduction — What is HieraSync AI?

Centralized departmental governance for academic institutes

CORE PLATFORM VISION

- **Unified Full-Stack Portal:**

Centralizes tasks, events, approvals, notifications, goals, and analytics.

- **Role-Aware Governance:**

Built around the institutional hierarchy (Principal → HOD → Faculty → Staff/Students).

- **10 Distinct Roles:**

Granular access rules for Principal, HOD, Faculty, TAs, Lab Assistants, Staff, Students, and Reps.

- **Single Source of Truth:**

Replaces paper registers, WhatsApp groups, and disconnected Excel sheets.



TECHNICAL & OPERATIONAL SCOPE

- **Domain:**

Academic Workflow Automation + Applied AI (heuristic risk scoring, summarization, reporting).

- **Target Environment:**

Designed for CSE (AI & ML) at SBJIT Nagpur, extensible to any college department.

- **Engineering Depth:**

17 FastAPI REST API routers, 13+ React UI pages, JWT security, and cron automation.

- **Cloud-Ready Architecture:**

React SPA + FastAPI backend + Firebase Firestore, Auth, and Storage.

Background & Operational Motivation

Addressing chronic friction in traditional academic administrative workflows

- **Motivation:**

digitize the hierarchy itself — every request flows through the correct authority with timestamps, comments and status.

- **Opportunity:**

cloud + AI can automate prioritization, reminders and delay-risk alerts at near-zero operational cost.

- **Outcome:**

faster decisions, fair workload distribution and accreditation-ready digital records (NAAC/NBA documentation).

2025–26 evidence: a UNSW pilot cut admin workload 60% with AI agents • 2026 surveys: Asana AI leads risk detection but paywalls AI at \$10.99+/user/mo — academic teams need a low-cost alternative.

Lost Approvals

Paper forms & chat messages leave no audit trail; requests stall for days without visibility.

Missed Deadlines

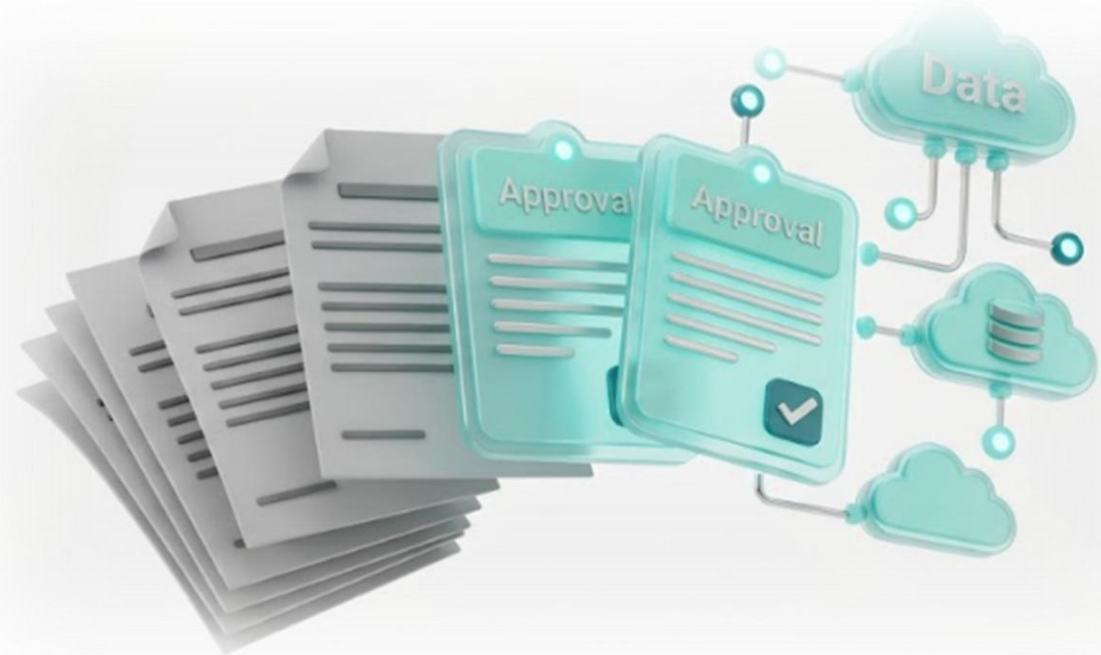
Lack of automated follow-ups and risk signals causes critical tasks to slip repeatedly.

Workload Opacity

HODs cannot track individual faculty load; task distribution relies on manual guesswork.

Manual Reporting

Monthly & NAAC/NBA reports are manually assembled from Excel sheets—slow and error-prone.

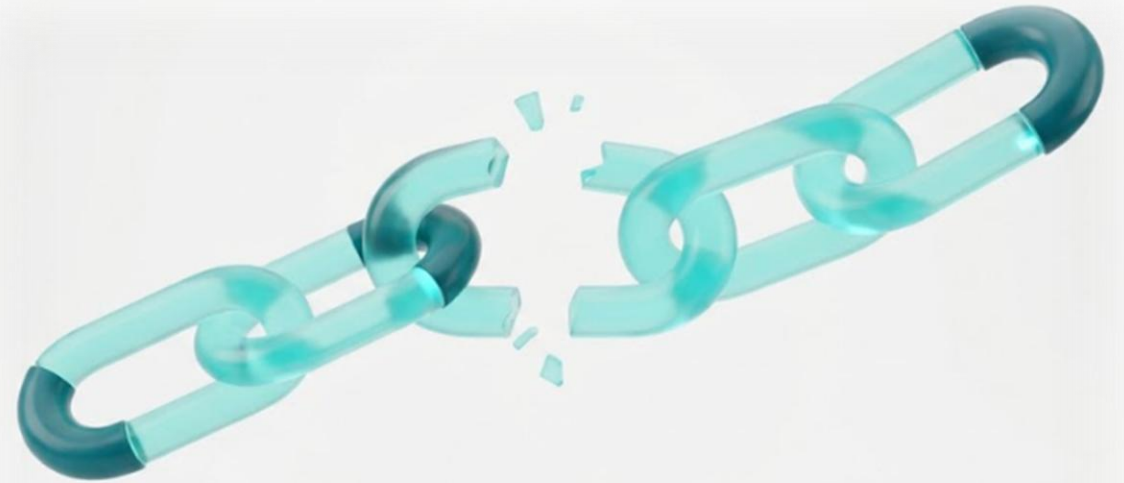


Problem Statement & Industry Context

The core operational bottleneck facing higher education departments

CORE PROBLEM STATEMENT

“Academic departments lack a unified, role-aware digital system to assign, track, approve, and analyse teaching and administrative work — causing delays, operational opacity, and severe workload imbalance.”



2025–26 BENCHMARK EVIDENCE

• UNSW Pilot Evidence:

A 2025 UNSW academic trial showed AI-assisted workflow agents cut administrative workload by 60%.

• Asana / Monday Surveys (2026):

Commercial PM suites lead in risk detection but paywall AI features at \$7–\$20/user/month—unaffordable for academic departments.

HIERASYNC TARGET SOLUTION

• Zero-Cost Architecture:

Eliminates per-seat licensing fees by building zero-cost heuristic AI risk scoring.

• Academic Hierarchy First:

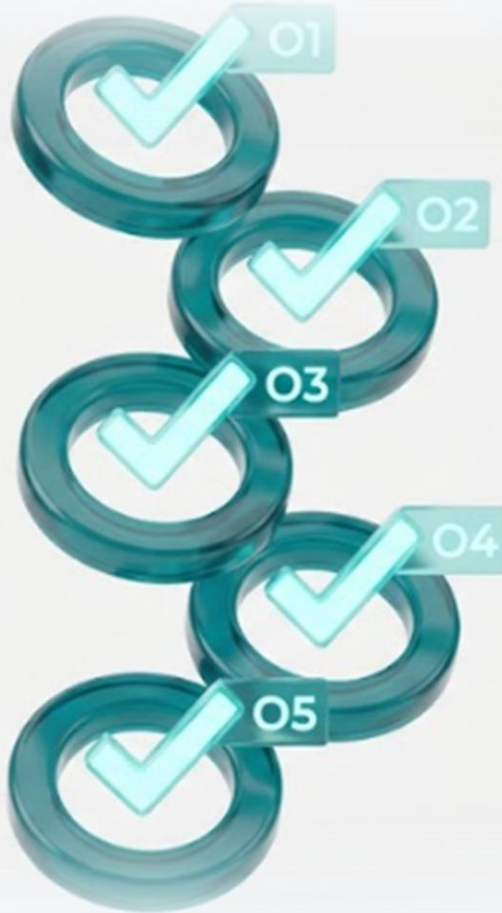
Implements exact institutional role inheritance (Principal → HOD → Faculty → Staff) rather than generic corporate flat structures.

Project Objectives & Scope

Defined outcomes and review boundaries

Scope:

Department-level web deployment: CSE (AI & ML). Web-first responsive UI; Firebase cloud backend; heuristic AI (no GPU servers).



Centralized Workspace

Consolidate tasks, events, approvals, documents, goals, and notifications into one secure portal.

Two-Stage Digital Approvals

Implement structured HOD → Principal workflow with comments, decision logs, and status tracking.

Follow-Up Automation

Deploy an in-app notification center, scheduled daily 8 AM reminders, and deadline alerts.

AI Assistance & Risk Engine

Integrate deadline-risk prediction, priority scoring, conversational assistant, and report generator.

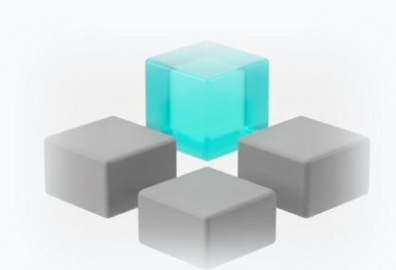
Analytics & One-Click Reports

Provide interactive dashboards, faculty performance views, and exportable department summaries.

Successive Criteria: login as each role → assign → approve → notify → analyse.

Existing Systems — Comparative Matrix

Evaluating current academic and commercial management solutions



System Category	Key Strengths	Identified Limitations	HieraSync Advantage
Manual Registers / Excel / WhatsApp	Zero monetary cost; highly familiar.	No tracking or audit trail; zero analytics; lost data.	Full digital audit log & central DB.
Traditional College ERPs	Handles fees, admissions, and records.	Costly, rigid, weak task workflows; poor faculty UX.	Lightweight, modular, web-first SPA.
Commercial AI PM Suites (Asana/Monday)	Rich risk detection & automation.	AI paywalled at \$7–20/user/mo; no academic hierarchy.	Built-in academic RBAC; zero AI cost.
Google Workspace / LMS (Moodle)	Strong collaboration & course content.	Scattered tools; no multi-stage approval workflows.	Unified approvals + task risk engine.

Inference (incl. 2026 app surveys): nothing combines academic hierarchy + workflows + approvals + analytics + AI in one low-cost stack — the gap HieraSync targets.

Research Papers & Identified Architectural Gaps

Grounded analysis of academic literature shaping HieraSync AI

IEEE Xplore (Doc 10743425) — Analysis of College ERP Software Systems

- Finding: ERPs raise efficiency, but cost and complexity block adoption.
- HieraSync Answer: Low-cost modular serverless cloud stack.

IJERT (Swain et al. 2026) — Modular Web-Based College ERP

- Finding: Cuts administrative labor through modular API slices.
- HieraSync Answer: Router-per-module FastAPI architecture.

Frontiers Educ. (Khairullah 2025) — AI in Higher Education Administration

- Finding: AI decision support optimizes resource allocation.
- HieraSync Answer: Heuristic risk engine + analytical dashboards.

Frontiers Educ. (Buele 2025) — Faculty Perceptions & Transformations

- Finding: Unguided AI overloads staff without structure.
- HieraSync Answer: Human-in-the-loop: AI alerts, humans decide.

GAPS → OUR ANSWER

- Prior work = student records & fees → we target faculty workflows
- AI paywalled \$7–20/user/mo → our zero-cost heuristic engine
- Full automation risks overload → human-in-the-loop approvals
- Heavy on-premise ERPs → serverless Firebase, web-only

Design Approach & Methodology

Agile, module-wise engineering slice strategy

DEVELOPMENT METHODOLOGY

- **Agile Module Slices:**

Each feature developed as an independent slice (API router + UI page + Firestore schema).

- **3-Tier Decoupling:**

Strict separation of Client SPA, FastAPI REST logic, and Firebase Cloud Data.

- **Security-First Middleware:**

Firebase Auth + JWT token validation; role dependency check on every endpoint.

- **Automation Layer:**

APScheduler cron background jobs for daily 8 AM deadline reminders.

KEY ARCHITECTURAL DECISIONS

- **Firestore NoSQL:**

Flexible workflow documents, serverless auto-scaling, zero DB management ops.

- **Router-per-Module API:**

17 independent FastAPI routers; easily extensible without code breakage.

- **Zero-Cost Heuristic AI:**

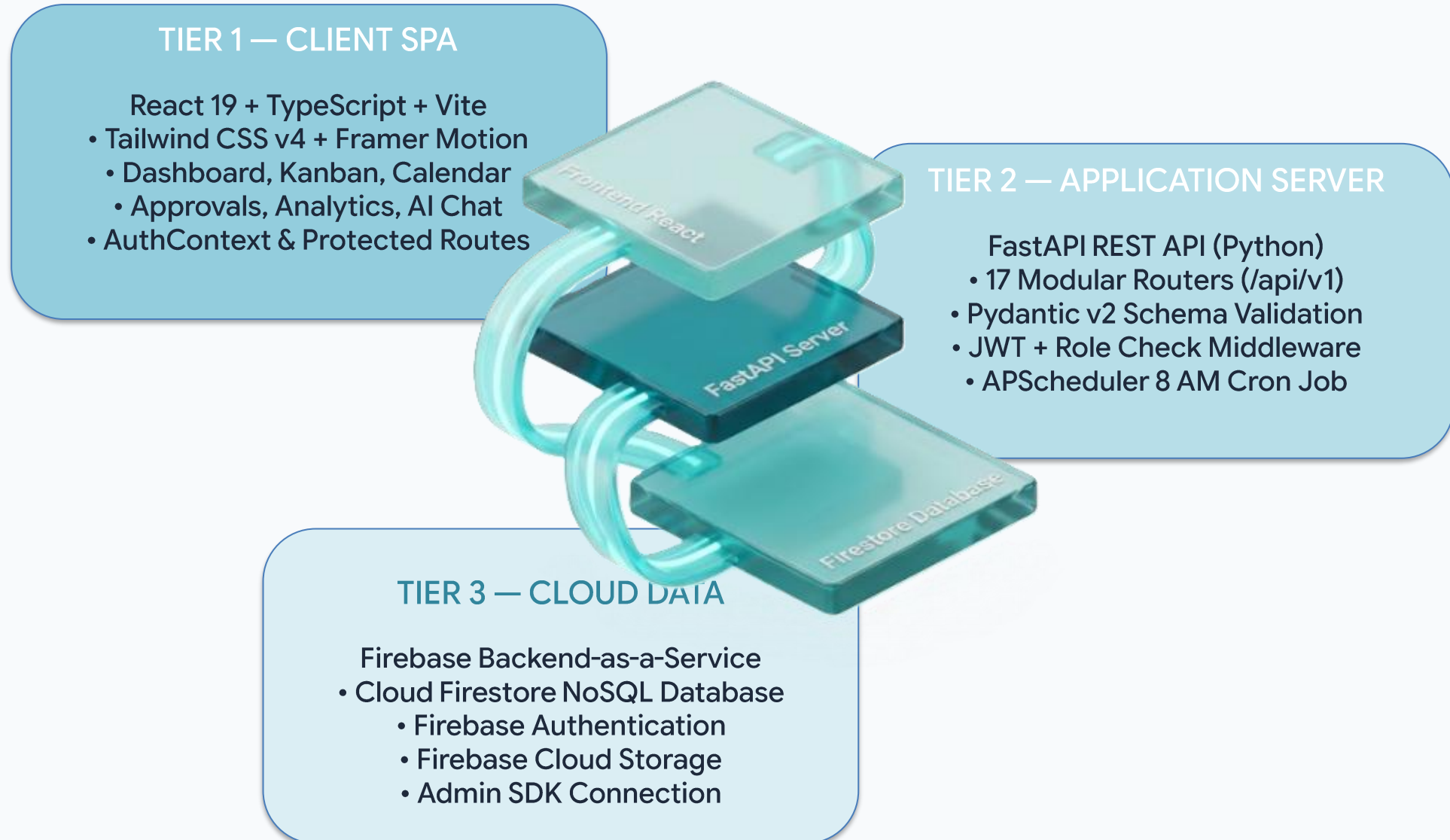
Deterministic risk scoring without GPU or LLM API server costs.

- **Audit-Friendly Approvals:**

Automatic timestamps, comments, and stage history logs on every action.

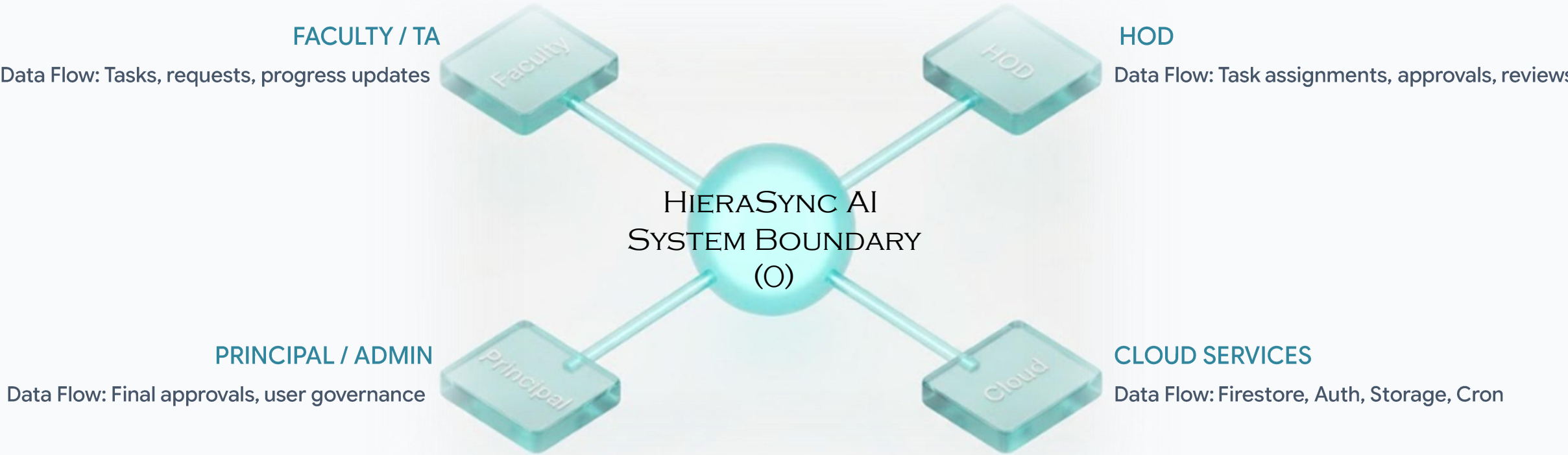
System Architecture (3-Tier Cloud Stack)

Decoupled web architecture powering HieraSync AI



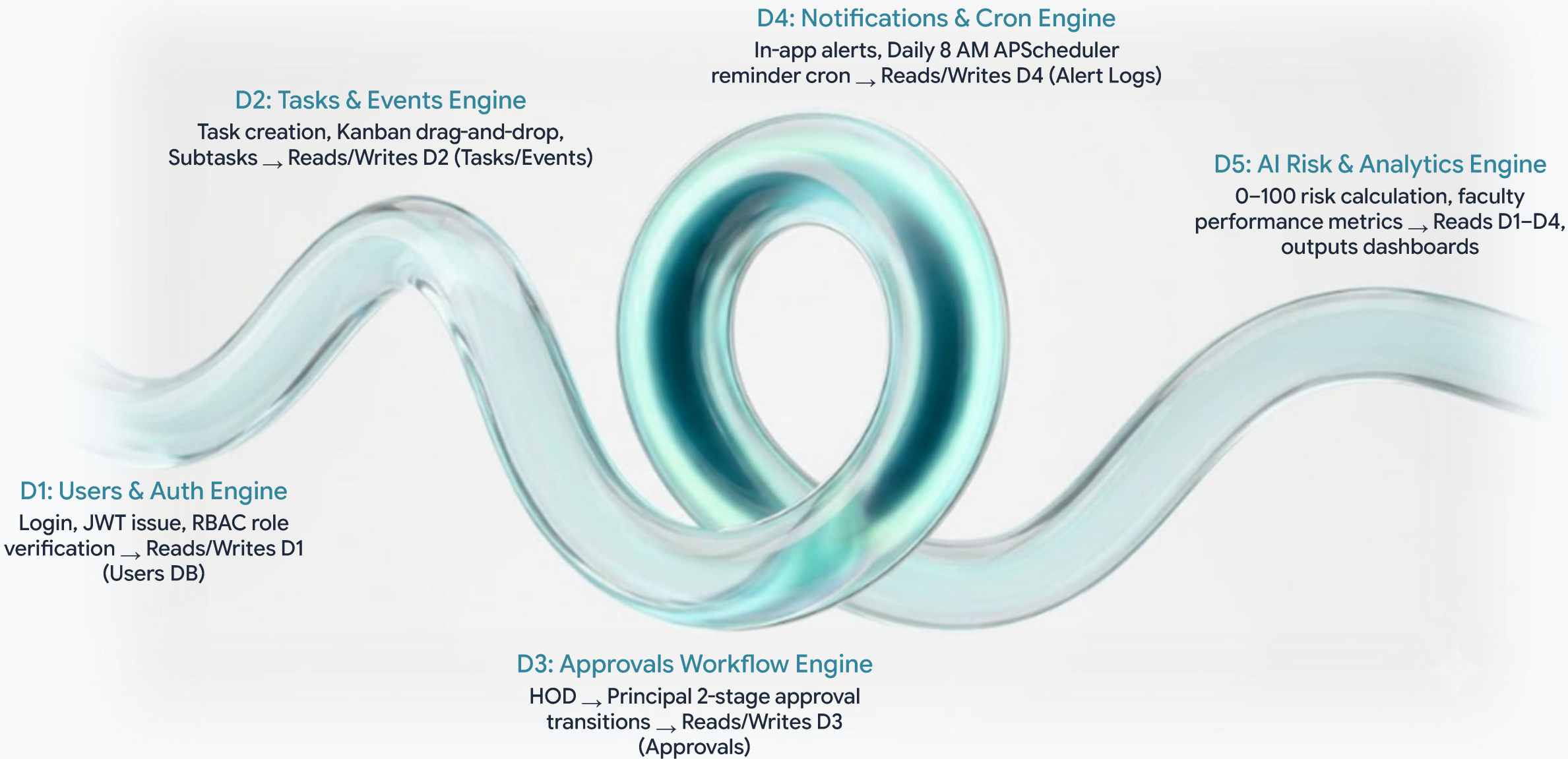
Data Flow Diagram — Level 0 (Context DFD)

High-level system boundary and external entity interactions



Data Flow Diagram — Level 1 (Detailed Subsystems)

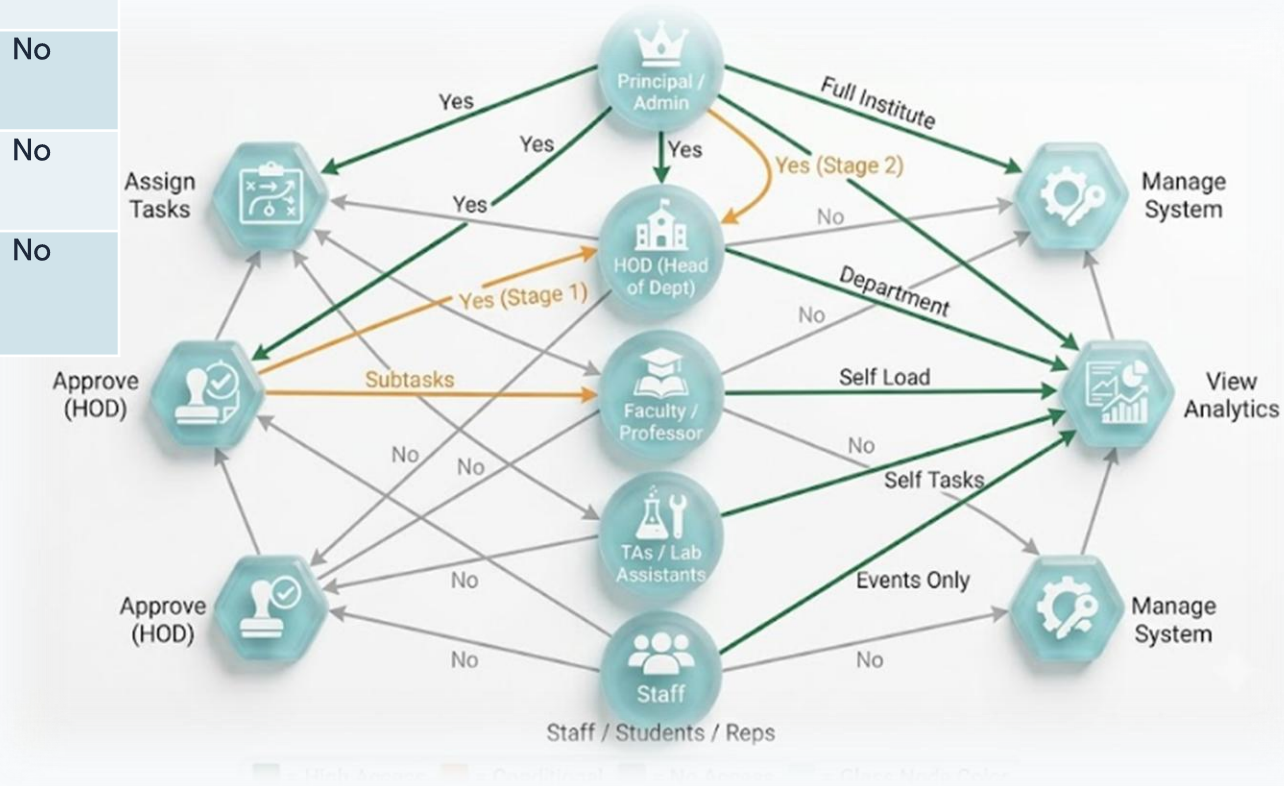
Interconnected functional process engines and Firestore datastores



Use Case Diagram & Granular RBAC Permissions

Role-based access matrix across 10 academic user levels

User Role	Assign Tasks	Approve (HOD)	Approve (Principal)	View Analytics	Manage System
Principal / Admin	Yes	Yes	Yes (Stage 2)	Full Institute	Yes
HOD (Head of Dept)	Yes	Yes (Stage 1)	No	Department	No
Faculty / Professor	Subtasks	No	No	Self Load	No
TAs / Lab Assistants	No	No	No	Self Tasks	No
Staff / Students / Reps	No	No	No	Events Only	No



Database Design — Cloud Firestore Schema

NoSQL collection structure, document schemas, and subcollections

users

id, name, email, role (10 roles), department_id, designation, status

tasks

id, title, assignee_id, priority, status, progress_pct, deadline, risk_score, goal_id

events

id, title, date, event_type, organizer_id, location

goals

id, title, category, target_date, milestones[]

departments

id, name, code, hod_id, created_at

approvals

id, title, requester_id, stage, hod_comment, principal_comment, status

notifications

id, user_id, title, message, is_read, created_at

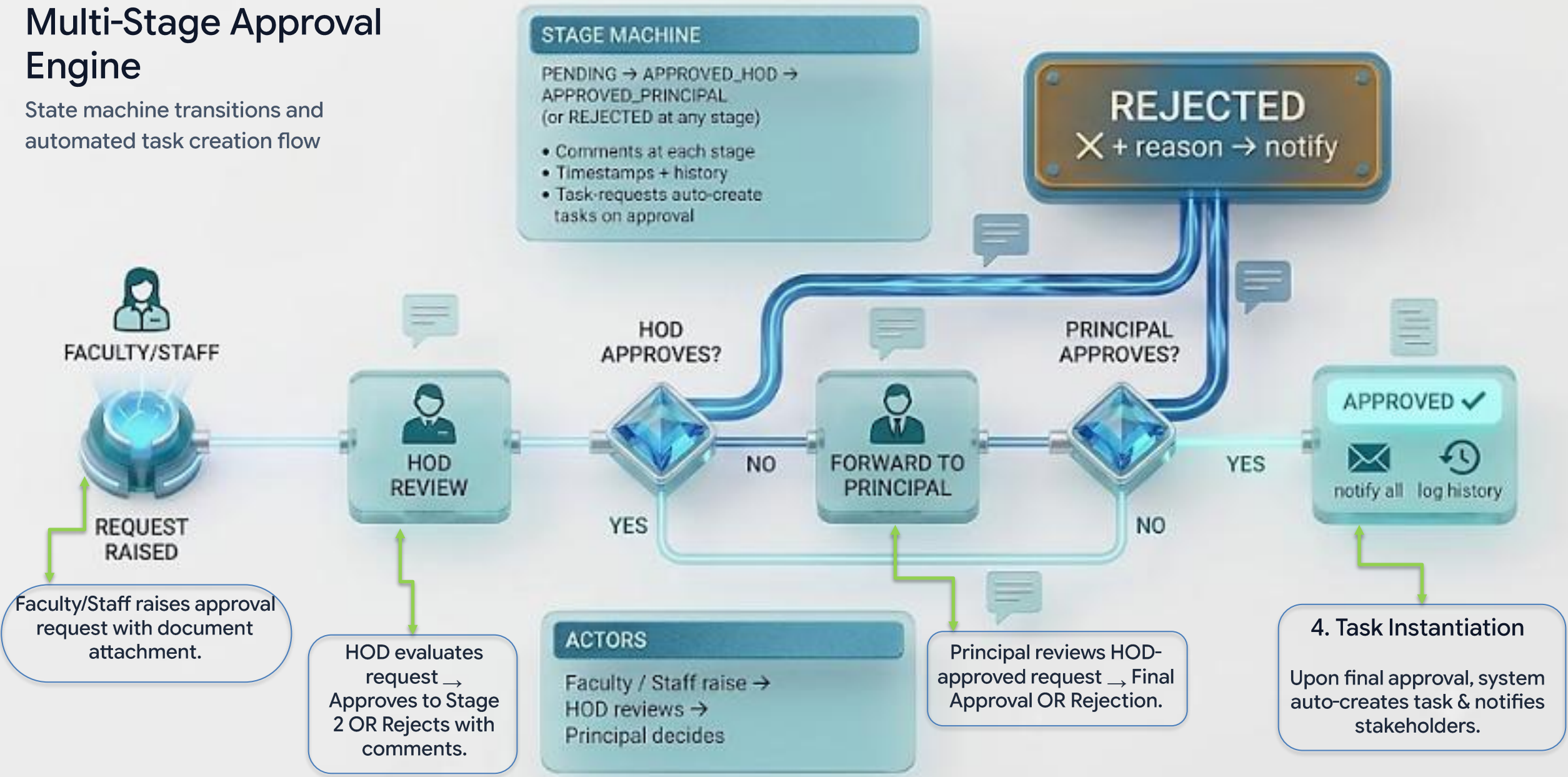
subcollections

comments/ (task comments, mentions) & attachments/ (files)



Core Workflow — Multi-Stage Approval Engine

State machine transitions and automated task creation flow



Technology Stack & Engineering Justifications

Production-grade open-source choices powering HieraSync AI

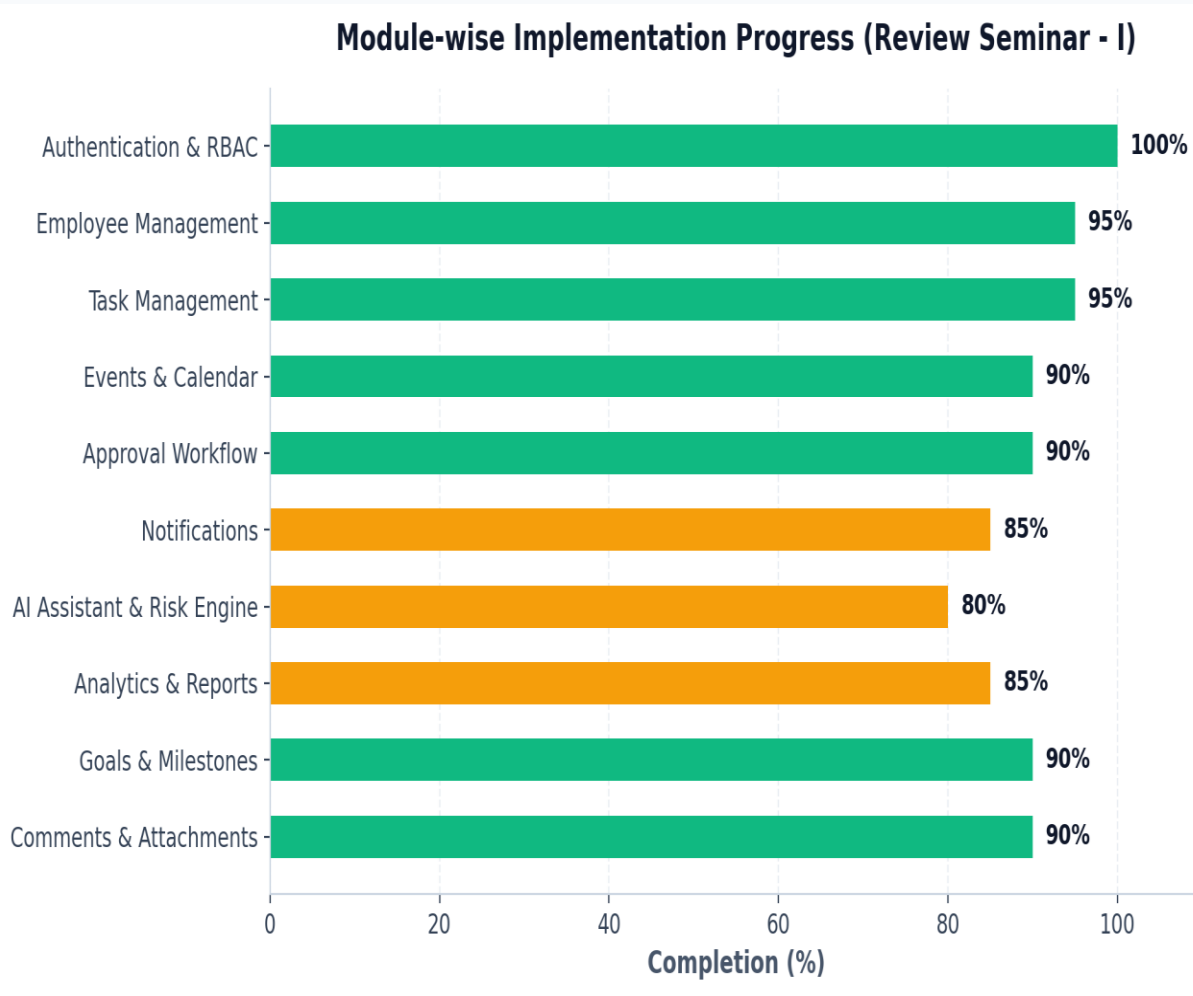


Layer	Selected Technology	Engineering Justification
Frontend SPA	React 19 + TypeScript + Vite	Actions/Suspense, strict static typing, instant HMR build times.
Styling & UI	Tailwind CSS v4 + Framer Motion	Utility-first responsive design system, smooth fluid animations.
Interactive UI	FullCalendar, Recharts, dnd-kit	Production-grade calendar, analytics charts, and drag-and-drop Kanban.
Backend REST API	FastAPI (Python) + Pydantic v2	Asynchronous performance, strict input validation, auto OpenAPI docs.
Security / Auth	Firebase Auth + JWT (python-jose)	Stateless, secure token authentication with per-request role enforcement.
Cloud Data / Cron	Cloud Firestore + APScheduler	Serverless NoSQL database scaling with background daily 8 AM reminder cron.

Modules Developed — Implementation Status (Review Seminar - I)

Current implementation state across all 9 core platform modules

#	Module	Key capability delivered	Status	
M1	Auth & RBAC	Login/register, JWT, 10 roles, protected routes	Done 100%	✓
M2	Employee Mgmt	Faculty CRUD, profiles, department mapping	Done 95%	✓
M3	Task Management	Kanban, subtasks, priorities, progress, risk badges	Done 95%	✓
M4	Events & Calendar	FullCalendar views, types, reminders	Done 90%	✓
M5	Approval Workflow	HOD→Principal stages, comments, history	Done 90%	✓
M6	Notifications	Center, unread counts, scheduler + email hooks	85%	🕒
M7	AI Assistant	Chat, dashboard summary, report-gen, risk engine	80%	🕒
M8	Analytics & Reports	Stats, activity log, faculty performance, export	85%	🕒
M9	Goals / Requests / Social	Milestones, task-requests, comments, attachments	Done 90%	✓



Deep-Dive A — Tasks & Heuristic AI Risk Engine

Interactive Kanban task management paired with deterministic risk prediction

TASK ENGINE & AI RISK CALCULATOR

- Kanban Board:

Drag-and-drop interface powered by dnd-kit with subtasks and progress bars.

- Goal Linking:

Connects individual faculty tasks directly to departmental goals.

- Deterministic Risk Math:

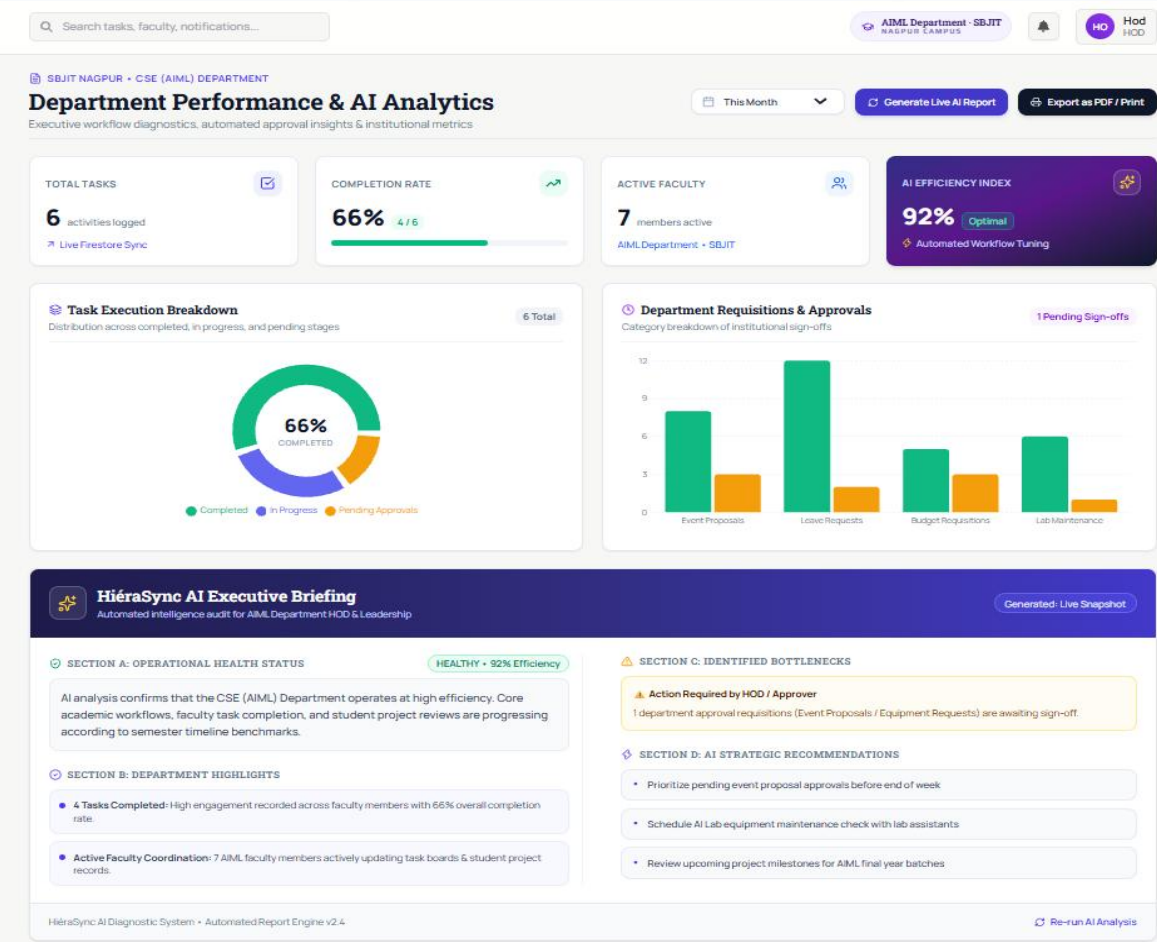
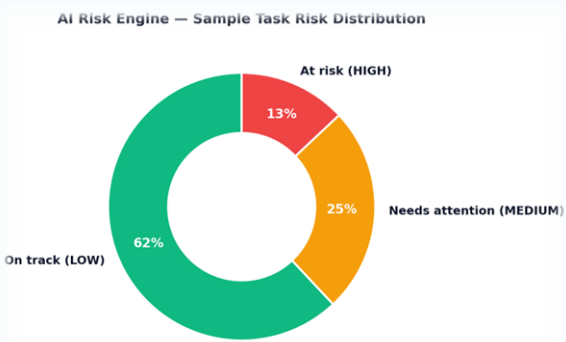
Evaluates risk score (0–100) using deadline gap, progress lag, and workload factor.

- Risk Levels:

Categorizes tasks into LOW, MEDIUM, and HIGH risk badges with explainable human-readable factors.

- Zero Operational Cost:

Runs instantly on FastAPI without requiring GPU servers or LLM API tokens.



Deep-Dive B — Approvals, Analytics, and Notifications

Multi-stage approval tracking, performance analytics, and cron automation

MULTI-STAGE APPROVALS

- Stage 1: HOD Review with comments
 - Stage 2: Principal Final Approval
- Complete timestamped audit history
- Task requests auto-convert to tasks upon approval.

ANALYTICS & REPORTS

- Departmental completion stats
 - Faculty workload & performance table
 - On-time task completion rate tracking
- One-click exportable PDF/Excel summaries.

NOTIFICATIONS & CRON

- In-app notification center with unread badge
 - Real-time alerts for approvals & tasks
- APScheduler background daily 8 AM cron job
 - Prepared email gateway hooks.

Advantages & Administrative ROI Analysis

Quantifiable reduction in institutional effort and processing time

CORE OPERATIONAL ADVANTAGES

- **Single Source of Truth:**

Replaces registers, emails, and WhatsApp threads with a single accountable portal.

- **Accelerated Decisions:**

Approval cycle time drops from days to hours with digital multi-stage routing.

- **Full Accountability:**

Timestamps, comments, and decision logs on every single approval action.

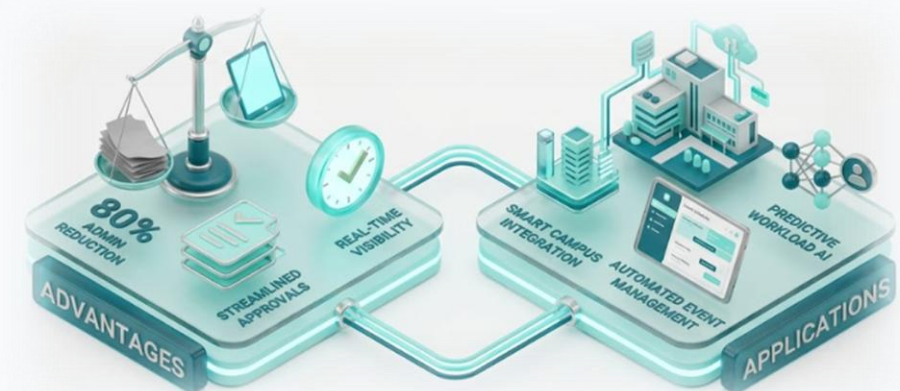
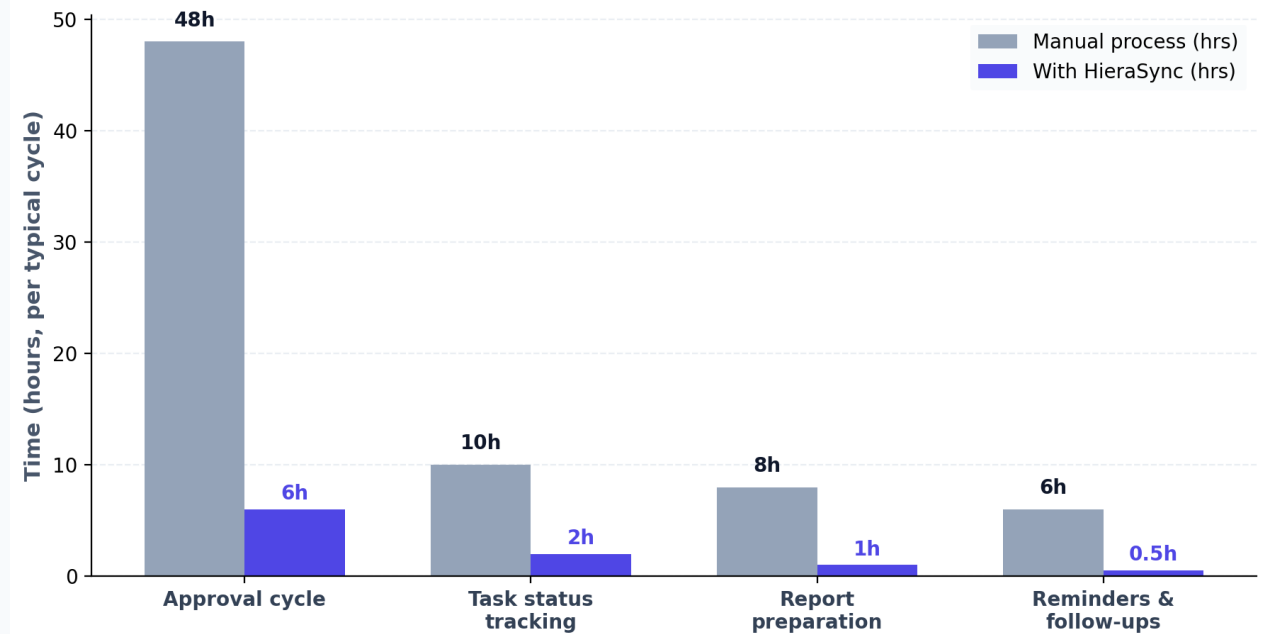
- **Proactive Delay Prevention:**

AI risk scoring flags overdue tasks before they impact department performance.

- **NAAC/NBA Readiness:**

One-click exportable reports simplify accreditation documentation.

Estimated Administrative Effort: Manual vs HieraSync



Applications & Future Expansion Roadmap

Target institutional deployment areas and long-term development scope



CURRENT APPLICATION AREAS

- **Department Task Governance:**

Assigning and tracking teaching, lab, and committee work.

- **Institutional Approvals:**

Leave requests, purchase requisitions, and hall/equipment bookings.

- **Accreditation Support:**

Instant data aggregation for NAAC, NBA, and NIRF annual reports.

- **Faculty Load Balancing:**

Evaluating real-time workload across department members.

- **Student Clubs & Events:**

Approving student event proposals and managing event calendars.

FUTURE EXPANSION SCOPE

- **Mobile PWA & Push:**

Progressive Web App with native mobile push notifications.

- **Machine Learning Models:**

Predictive ML models trained on task history to forecast delays.

- **ERP/LMS Integration:**

Syncing with college attendance and timetable databases.

- **WhatsApp/SMS Gateway:**

Automated SMS/WhatsApp notification triggers for urgent tasks.

- **Multi-College SaaS:**

Multi-tenant architecture for institute-wide console deployment.

References & Academic Bibliography

Citations of research papers, journals, standards, and repositories

- 1] “A Detailed Analysis of College ERP Software Systems,” IEEE Xplore, doc. 10743425.
<https://ieeexplore.ieee.org/document/10743425/>
- 2] S. Swain et al., “College ERP Management System,” IJERT, vol. 15, no. 04, Apr. 2026, DOI: 10.5281/zenodo.19731754.
- 3] S. Tripathi et al., “College ERP System,” IJCRT, paper IJCRT1812344.
<https://www.ijcrt.org/papers/IJCRT1812344.pdf>
- 4] R. Salunke et al., “ERP Management System” (Django, RBAC), IJARSCT.
<https://www.ijarsct.co.in/Paper25814.pdf>
- 5] S. A. Khairullah et al., “Implementing AI in academic and administrative processes...,” Front. Educ., 2025.
<https://www.frontiersin.org/journals/education/articles/10.3389/feduc.2025.1548104/full>
- 6] “The Future of AI Agents in Higher Education,” ResearchGate, 2025.
<https://www.researchgate.net/publication/394249334>
- 7] “Automating Higher Education Administrative Processes with AI-Powered Workflows,” ResearchGate.
<https://www.researchgate.net/publication/399184231>
- 8] J. Buele et al., “Transformations in academic work and faculty perceptions of AI,” Front. Educ., 2025.
<https://www.frontiersin.org/journals/education/articles/10.3389/feduc.2025.1603763/full>
- 9] “8 Best AI for Project Management in 2026 (Compared by a PM),” Duple, 2026.
<https://duple.com/learn/best-ai-for-project-management>
- 10] FastAPI, Firebase, React & Vite official docs; Project repo:
https://github.com/Parthwadekar40/Hiera_Sync

Conclusion & Key Takeaways

SUMMARY OF DELIVERABLES

- Unified Role-Aware System:

Built a functional full-stack web platform covering departmental workflows end-to-end.

- Digitized Approvals & Follow-ups:

Multi-stage HOD → Principal routing with automatic reminders prevents stalled tasks.

- Zero-Cost AI Foresight:

Heuristic risk scoring provides proactive delay alerts without per-seat AI subscription costs.

- Runnable from Repository:

All 5 core objectives are implemented and runnable directly from the repository today.

“ HieraSync turns departmental chaos into a calm, transparent workflow. ”

Thank You!